1 System Structure

1.1 Unbranched Reaction Chain

$$X_1 \xrightarrow{v_1} S_1 \xrightarrow{v_2} S_2 \cdots S_{n-1} \xrightarrow{v_n} S_n \xrightarrow{v_{n+1}} X_2$$

- X_1 = external substrate (constant)
- X_2 = external product (constant)
- $S_1, \ldots, S_n$ = internal metabolites
- $v_1, \ldots, v_{n+1}$ = reaction rates
- $r = n + 1$ = number of reactions

1.2 Steady-State Property

At steady state, all reaction rates equal the flux:

$$v_j = J \quad \text{for } j = 1, \ldots, n+1$$

Key consequence: The flux control coefficient matrix has identical rows:

$$C_j^J = C_{ij}^J \quad \text{for all } i$$

2 Linear Kinetics (Non-Saturated)

2.1 Conditions for Linear Regime

When enzymes operate far from saturation:

$$J \ll V_j^f, V_j^r$$
$$S_i \ll K_i^r, K_{i+1}^f$$

V_j^f is the forward maximal rate and V_j^r the reverse maximal rate.

K_i^r is the product K_m for the producing step i, and K_{i+1}^f is the K_m for the substrate of the consuming step.

Under these conditions we can approximate rate laws using a linear rate law.

2.2 Linear Rate Law

$$v_j = k_j S_{j-1} - k_{-j} S_j = k_j \left(S_{j-1} - \frac{S_j}{q_j} \right)$$

where:

$$k_j = \frac{V_j^f}{K_j^f} \quad \text{(forward rate constant)}$$

$$k_{-j} = \frac{V_j^r}{K_j^r} \quad \text{(backward rate constant)}$$

$$q_j = \frac{k_j}{k_{-j}} = \frac{V_j^f K_j^r}{V_j^r K_j^f} \quad \text{(equilibrium constant)}$$

The last relation is the **Haldane relationship**.

2.3 Metabolite Concentrations

$$S_i = P_1 \prod_{j=1}^{i} q_j - J \sum_{l=1}^{i} \frac{1}{k_l} \prod_{j=l}^{i} q_j$$

2.4 Steady-State Flux

$$J = \frac{X_1 \prod_{j=1}^{n+1} q_j - X_2}{\sum_{l=1}^{n+1} \frac{1}{k_l} \prod_{j=l}^{n+1} q_j}$$

Interpretation:

- Numerator: thermodynamic driving force
- Denominator: sum of resistances weighted by downstream equilibria

3 Flux Control Coefficients

3.1 Definition via Rate Constants

$$C_j^J = \frac{k_j}{J} \frac{\partial J}{\partial k_j} = \frac{v_j}{J} \frac{\partial J / \partial k_j}{\partial v_j / \partial k_j}$$

(Perturbation keeps equilibrium constants fixed)

3.2 Explicit Formula (Linear Kinetics)

$$C_i^J = \frac{\frac{1}{k_i} \prod_{j=i}^{n+1} q_j}{\sum_{l=1}^{n+1} \frac{1}{k_l} \prod_{j=l}^{n+1} q_j}$$

3.3 Fundamental Properties

$$0 \le C_j^J \le 1 \quad \text{(non-negative, bounded)}$$

$$\sum_{j=1}^{n+1} C_j^J = 1 \quad \text{(summation theorem)}$$

Note: Normalization doesn't change values since $v_j = J$ at steady state.

3.4 Ratio of Control Coefficients

$$\frac{C_j^J}{C_i^J} = \frac{k_i}{k_j} \prod_{l=j}^{i-1} q_l \quad \text{with } i > j$$

Key insight: If reaction s is irreversible ($q_s \to \infty$), then $C_i^J \to 0$ for all reactions $i > s$ downstream.

3.5 Interpretation

- Fast reactions (k_j large) $\Rightarrow$ low C_j^J
- Slow reactions (k_j small) $\Rightarrow$ high C_j^J
- Position in chain matters via $\prod q_j$ terms
- Irreversible steps "shield" downstream reactions

4 Finite Perturbations

4.1 Single Reaction Perturbation

$$\frac{\Delta J}{J} = \frac{C_k^J \, \Delta v_k / v_k}{1 + (1 - C_k^J) \, \Delta v_k / v_k}$$

This exact result applies to unbranched chains with linear kinetics.

4.2 Multiple Perturbations

$$\frac{\Delta J}{J} = \sum_{j=1}^{n+1} C_j^J \left(\frac{\Delta v_j / v_j}{1 + \Delta v_j / v_j} \right) \left(\sum_{j=1}^{n+1} \frac{C_j^J}{1 + \Delta v_j / v_j} \right)^{-1}$$

4.3 Linear Approximation

For small perturbations ($|\Delta v_j / v_j| \ll 1$):

$$\frac{\Delta J}{J} \approx \sum_{j=1}^{n+1} C_j^J \frac{\Delta v_j}{v_j}$$

5 General Treatment (Nonlinear)

5.1 Elasticity Structure

For unbranched chains where v_i depends only on S_{i-1} (substrate) and S_i (product):

$$\varepsilon_{ij} = \varepsilon_{ii}\, \delta_{ij} + \varepsilon_{i,i-1}\, \delta_{i-1,j}$$

Typical signs:

$$\begin{aligned} \varepsilon_{ii} &< 0 \quad \text{(product inhibition)} \\ \varepsilon_{i+1,i} &> 0 \quad \text{(substrate activation)} \end{aligned}$$

5.2 Two-Enzyme System

$$X_1 \xrightarrow{v_1} S_1 \xrightarrow{v_2} X_2$$

Flux control:

$$C_1^J = \frac{\varepsilon_{21}}{\varepsilon_{21} - \varepsilon_{11}}$$
$$C_2^J = \frac{-\varepsilon_{11}}{\varepsilon_{21} - \varepsilon_{11}}$$

Concentration control:

$$C_{11}^S = \frac{1}{\varepsilon_{21} - \varepsilon_{11}}$$
$$C_{12}^S = \frac{-1}{\varepsilon_{21} - \varepsilon_{11}}$$

5.3 Connectivity Theorem (Unbranched)

$$C_i^J \varepsilon_{ii} + C_{i+1}^J \varepsilon_{i+1,i} = 0$$

Rearranged:

$$\frac{C_i^J}{C_{i+1}^J} = -\frac{\varepsilon_{i+1,i}}{\varepsilon_{ii}}$$

5.4 General Flux Control Formula

$$C_i^J = \frac{\prod_{j=i}^{n} \left(-\frac{\varepsilon_{j+1,j}}{\varepsilon_{jj}} \right)}{1 + \sum_{j=1}^{n} \prod_{l=j}^{n} \left(-\frac{\varepsilon_{l+1,l}}{\varepsilon_{ll}} \right)}$$

When $\varepsilon_{ii} < 0$ and $\varepsilon_{i+1,i} > 0$, all $C_j^J \geq 0$.

5.5 Implications for Enzyme Properties

- **Near-equilibrium enzymes:** High elasticities $\Rightarrow$ low flux control
- **Substrate-saturated enzymes:** Low $\varepsilon_{i,i-1} \Rightarrow$ high C_{i-1}^J
- **Product-saturated enzymes:** Low $|\varepsilon_{ii}| \Rightarrow$ low C_{i+1}^J

6 Concentration Control Coefficients

6.1 Connectivity Theorem

$$C_{ij}^S \varepsilon_{jj} + C_{i,j+1}^S \varepsilon_{j+1,j} = -\delta_{ij}$$

6.2 Recurrence Relations

For $j \neq i$:

$$C_{ij}^S = -C_{i,j+1}^S \frac{\varepsilon_{j+1,j}}{\varepsilon_{jj}} \qquad \text{for } 1 \leq j < i$$

$$C_{i,j+1}^S = -C_{ij}^S \frac{\varepsilon_{jj}}{\varepsilon_{j+1,j}} \qquad \text{for } i+1 \leq j \leq n$$

6.3 Relation to Flux Control

$$C_{ij}^S = C_{ii}^S \frac{C_j^J}{C_i^J} \qquad \text{for } 1 \leq j \leq i$$

$$C_{ij}^S = C_{i,i+1}^S \frac{C_j^J}{C_{i+1}^J} \qquad \text{for } i+1 \leq j \leq n+1$$

6.4 Final Formulas

$$C_{ij}^S = \frac{C_j^J}{C_{i+1}^J \varepsilon_{i+1,i}} \sum_{k=i+1}^{n+1} C_k^J \qquad \text{for } j \leq i$$

$$C_{ij}^S = \frac{C_j^J}{C_i^J \varepsilon_{ii}} \sum_{k=1}^{i} C_k^J \qquad \text{for } j > i$$

6.5 Sign Patterns (Crossover Theorem)

When $\varepsilon_{ii} < 0$ and $\varepsilon_{i+1,i} > 0$:

Condition	Effect
$j \leq i$ (upstream)	$C_{ij}^S > 0$
$j > i$ (downstream)	$C_{ij}^S < 0$

Interpretation: Activating an enzyme:

- **Decreases** concentrations of upstream metabolites
- **Increases** concentrations of downstream metabolites

6.6 Special Cases

- $C_j^J \approx 0$ (fast enzyme) $\Rightarrow C_{ij}^S \approx 0$
- All downstream enzymes fast $\Rightarrow S_i$ in quasi-equilibrium with P_2
- All upstream enzymes fast $\Rightarrow S_i$ in quasi-equilibrium with P_1

7 Feedback Inhibition

7.1 System Structure

$$X_1 \xrightarrow{v_1} S_1 \longrightarrow \cdots \longrightarrow S_n \xrightarrow{v_{n+1}} X_2$$

feedback inhibition

Assumptions:

- All reactions irreversible: $\varepsilon_{ii} = 0$
- S_n inhibits v_1: $\varepsilon_{1,n} < 0$

7.2 Flux Control Coefficients

Internal enzymes have **zero** flux control:

$$C_j^J = 0 \quad \text{for } j = 2, \ldots, n$$

Only E_1 and E_{n+1} share flux control:

$$C_1^J = \frac{\varepsilon_{n+1,n}}{\varepsilon_{n+1,n} - \varepsilon_{1,n}}, \quad C_{n+1}^J = \frac{-\varepsilon_{1,n}}{\varepsilon_{n+1,n} - \varepsilon_{1,n}}$$

7.3 Control Ratio

$$\frac{C_{n+1}^J}{C_1^J} = -\frac{\varepsilon_{1,n}}{\varepsilon_{n+1,n}}$$

Without feedback ($\varepsilon_{1,n} = 0$): $C_1^J = 1$, $C_{n+1}^J = 0$

Strong feedback ($|\varepsilon_{1,n}| \gg |\varepsilon_{n+1,n}|$): Control shifts to E_{n+1}

7.4 Concentration Control

First enzyme:

$$C_{i,1}^S = \frac{C_1^J}{\varepsilon_{i+1,i}} > 0 \quad (i = 1, \ldots, n)$$

Last enzyme:

$$C_{i,n+1}^S = \begin{cases} \dfrac{C_{n+1}^J}{\varepsilon_{i+1,i}} > 0 & \text{for } i \neq n \\ -\dfrac{1}{\varepsilon_{n+1,n} - \varepsilon_{1,n}} < 0 & \text{for } i = n \end{cases}$$

Internal enzymes ($j = 2, \ldots, n$):

$$C_{i,i+1}^S = -\frac{1}{\varepsilon_{i+1,i}} < 0$$

(Only control their own substrate concentration)

7.5 Homeostasis Effect

Strong feedback ($|\varepsilon_{1,n}| \gg 1$):

$$C_{i,1}^S \ll 1 \quad \text{(weak input control)}$$
$$|C_{n,n+1}^S| \ll 1 \quad \text{(robust } S_n \text{ concentration)}$$

This is the mathematical basis for homeostatic regulation in metabolic pathways.

8 Summary Table

Quantity	Linear	General
Flux J	Explicit (Eq. 5.88)	Implicit
C_j^J	$\dfrac{(1/k_j)\prod q}{\sum(1/k)\prod q}$	$\dfrac{\prod(-\varepsilon/\varepsilon)}{1+\sum\prod}$
C_{ij}^S	Via C_j^J	Via connectivity
$\sum C_j^J$	$= 1$	$= 1$
Sign of C_j^J	≥ 0 always	≥ 0 (typical)

9 Quick Reference

Key relationships:

1. $v_j = J$ at steady state (all rates equal)
2. $C_i^J / C_{i+1}^J = -\varepsilon_{i+1,i}/\varepsilon_{ii}$
3. Irreversible step shields downstream control
4. Feedback shifts control to output enzyme

Crossover theorem:

- Activate enzyme $\Rightarrow$ decrease upstream S
- Activate enzyme $\Rightarrow$ increase downstream S

Created by Herbert M. Sauro, Based on: Heinrich & Schuster, "The Regulation of Cellular Systems" (1996), Chapter 5.4.3, March 19, 2026